

Design Patterns

Rodrigo Martins Pagliares

Design Patterns

Introduction

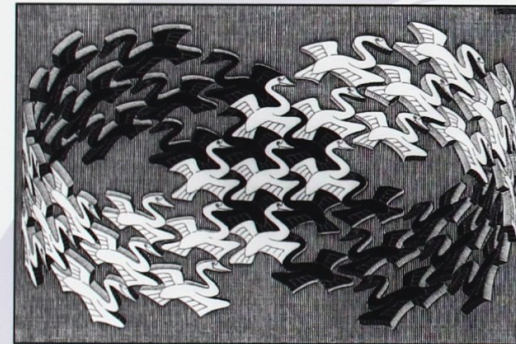
Introduction

- Course **overview**
- **Hands-on** course
- **Tools** used in this course
- **Design principles**
- What are **design patterns**?
- The **23 design patterns (GOF)**

Design Patterns

Elements of Reusable
Object-Oriented Software

Erich Gamma
Richard Helm
Ralph Johnson
John Vlissides



Cover art © 1994 M.C. Escher / Cordon Art - Baarn - Holland. All rights reserved.

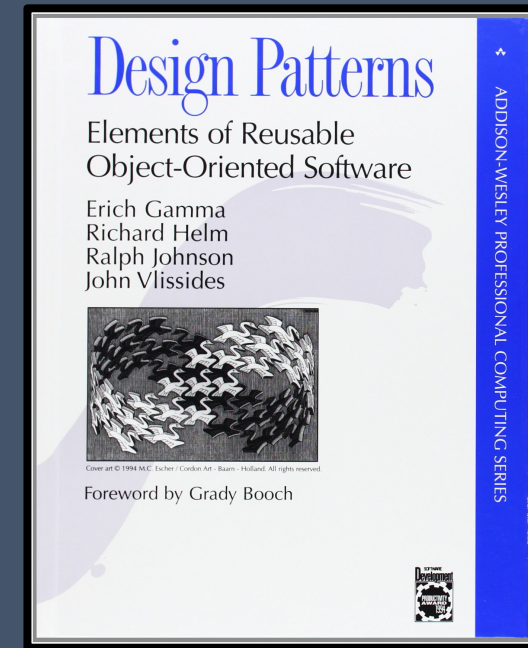
Foreword by Grady Booch



The 23 design patterns (GOF)

23 Design Patterns (GOF)

1. Factory method
2. Abstract factory
3. Builder
4. Prototype
5. Singleton
6. Adapter
7. Bridge
8. Composite
9. Memento
10. Observer
11. State
12. Strategy
13. Decorator
14. Facade
15. Flyweight
16. Proxy
17. Interpreter
18. Template method
19. Chain of responsibility
20. Command
21. Iterator
22. Mediator
23. Visitor



Tools used in this course

I use the IDE IntelliJ IDEA in this course.

IJ IntelliJ IDEA

Capable and Ergonomic IDE

[Download](#)

Why IntelliJ IDEA

Enjoy Productive Java

Every aspect of IntelliJ IDEA has been designed to maximize developer productivity. Together,

[X]
Cookies and IP addresses allow us to deliver and improve our web content and to provide you with a personalized experience. Our website uses cookies and collects your IP address for these purposes. [Learn more](#)

| JetBrains may use cookies and my IP address to |
| collect individual statistics and to provide me with |
| personalized offers and ads subject to the [Privacy](#) |
| [Policy](#) and the [Terms of Use](#). JetBrains may use |
| [third-party services](#) for this purpose. I can revoke |
| my consent at any time by visiting the [Opt-Out page](#). |
| Yes, I agree No, thanks |

<https://www.jetbrains.com/idea>

IntelliJ IDEA



Version: 2020.2.4
Build: 202.8194.7
24 November 2020

[Release notes](#) ➔

[System requirements](#)

[Installation Instructions](#)

[Other versions](#)

There are **2 editions (Community and Ultimate)**. The **community** version is **simpler and free** for all. The **Ultimate** is a **powerful IDE** that **enhances developer productivity** and **free** for **registered students and instructors**. You can use **anyone** in this course.

For web and enterprise development

Download

Free 30-day trial

For JVM and Android development

Download

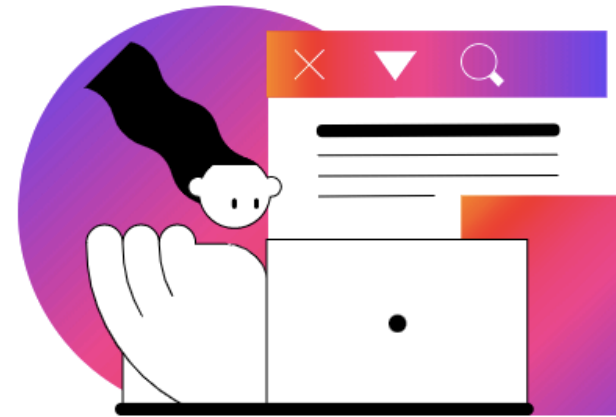
Free, open-source

	IntelliJ IDEA Ultimate	IntelliJ IDEA Community Edition ⓘ
Java, Kotlin, Groovy, Scala	✓	✓
Android ⓘ	✓	✓
Maven, Gradle, sbt	✓	✓
Git, SVN, Mercurial	✓	✓
Debugger	✓	✓
Profiling tools ⓘ	✓	✗
Spring, Java EE, Micronaut, Quarkus, Helidon, and more ⓘ	✓	✗

Free License Programs[Academic Licensing](#) [Open Source](#) [User Groups](#) [Events Partnership](#) [Developer Recognition](#)

Free Educational Licenses

Learn or teach coding with best-in-class development tools from JetBrains!



Students can get access to IntelliJ IDEA Ultimate for free by creating a JetBrains account.

Individual licenses for students and teachers

<https://www.jetbrains.com/community/education/#students>

Questions?